THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — SCIENCE · MEDIUM LEVEL FULL-YEAR PAPER SET 1

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The sum of coefficients in the balanced equation $\mathrm{Fe} + \mathrm{H_2O} \longrightarrow \mathrm{Fe_3O_4} + \mathrm{H_2}$ is: 1

(a) 8 **(b)** 9 **(c)** 10 **(d)** 12

2. The pH of a solution is 5. If its pH becomes 3, its $\mathrm{H^+}$ concentration: 1

(a) increases 100 times **(b)** decreases 2 times
(c) decreases 100 times **(d)** increases 2 times

3. Which of the following will not react with dilute HCl? 1

(a) zinc **(b)** magnesium **(c)** silver **(d)** iron

4. The number of covalent bonds in butane $\mathrm{C_4H_{10}}$ is: 1

(a) 14 **(b)** 10 **(c)** 12 **(d)** 13

5. In photosynthesis, the oxygen released comes from: 1

(a) glucose **(b)** water **(c)** chlorophyll **(d)** carbon dioxide

6. In a reflex arc, the correct order is: 1

(a) receptor -> sensory neuron -> spinal cord -> motor neuron -> effector
(b) receptor -> spinal cord -> brain -> effector
(c) effector -> motor neuron -> spinal cord -> sensory neuron -> receptor
(d) receptor -> motor neuron -> spinal cord -> sensory neuron -> effector

7. In which of the following does the offspring differ least from the parent? 1

(a) Offspring produced by fertilisation **(b)** Offspring produced by cross-pollination
(c) Offspring produced by budding in Hydra **(d)** Offspring produced by seeds

8. A heterozygous tall plant (Tt) is crossed with a dwarf plant (tt). The fraction of dwarf plants in the offspring is 1

(a) 3/4 **(b)** 1/2 **(c)** 1/4 **(d)** 0

9. An object is placed at the centre of curvature of a concave mirror of focal length 12 cm. The distance of the image from the mirror is 1

(a) 12 cm **(b)** 6 cm **(c)** 36 cm **(d)** 24 cm

10. A person with myopia has a far point at 100 cm. The lens needed has a power of 1

(a) -0.1 D **(b)** +10 D **(c)** -1 D **(d)** +1 D

11. The number of electrons passing per second through a conductor carrying 3.2 A is 1

(a) 2×10^{18} **(b)** 5×10^{18} **(c)** 3.2×10^{19} **(d)** 2×10^{19}

12. A compass needle placed below a wire carrying current from south to north is deflected. If the current is reversed, the deflection 1

(a) doubles **(b)** is reversed **(c)** becomes zero **(d)** stays the same

13. In the food chain Phytoplankton, Zooplankton, Small fish, Large fish the secondary consumer is 1

(a) phytoplankton **(b)** small fish **(c)** zooplankton **(d)** large fish

14. In $Fe_2O_3 + 2\,Al \longrightarrow Al_2O_3 + 2\,Fe$, the substance reduced is: 1

(a) Al_2O_3 **(b)** Al **(c)** Fe **(d)** Fe_2O_3

15. The solid residue obtained on heating washing soda crystals is: 1

(a) sodium bicarbonate **(b)** sodium chloride
(c) sodium hydroxide **(d)** anhydrous sodium carbonate

16. Which of the following is correct about ionic compounds? 1

(a) They are gaseous **(b)** They are insoluble in water
(c) They have high melting points **(d)** They are soft

17. CH_3CH_2OH on heating with excess conc. H_2SO_4 at 443 K gives: 1

(a) CH_3COOH **(b)** $CH_2{=}CH_2$ **(c)** CH_3CH_3 **(d)** CH_3CHO

18. Which has the longest small intestine? 1

(a) carnivores like tiger **(b)** humans
(c) birds **(d)** herbivores such as the cow

19. Why are shoots positively phototropic and roots negatively phototropic? 1

(a) roots need light
(b) shoots need gravity
(c) roots have no auxin
(d) auxin affects stem and root cells in opposite ways

20. The thread-like structure on which anther is held is the 1

(a) filament **(b)** ovule **(c)** stigma **(d)** style

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. Differentiate between a physical change and a chemical change with one example each. 2

22. Differentiate between a strong acid and a weak acid with one example each. 2

23. Why does calcium float while reacting with water? 2

24. Why does carbon form covalent compounds only and not ionic ones? 2

25. Why does a leaf kept in dark not show starch? How is this tested? 2

26. Why are reflexes important for survival? 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. Balance $Al + H_2SO_4 \longrightarrow Al_2(SO_4)_3 + H_2$ and state whether it is a displacement or a combination reaction. 3

28. A student adds Mg ribbon to three test tubes with HCl, CH_3COOH and distilled water. Predict the observation in each and explain. 3

29. Explain with equations how aluminium reacts with oxygen and what protects it from further corrosion. 3

30. Calculate the number of covalent bonds in 3
(a) hexane C_6H_{14} and
(b) $C_{10}H_{22}$.

31. Explain how the leaf is adapted for photosynthesis. 3

OR

A person's cerebrum is damaged. Which functions are affected? 3

32. Explain why a large number of spores are produced by Rhizopus even though only a few land in a suitable place. 3

OR

A tall pea plant is crossed with a dwarf and the F1 are all tall. When the F1 are self-pollinated, 300 plants result. Find the numbers of tall and dwarf plants and the genotypes. 3

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. Classify each as combination, decomposition, displacement or double displacement: 5
(a) $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$
(b) $2\,KClO_3 \longrightarrow 2\,KCl + 3\,O_2$
(c) $BaCl_2 + H_2SO_4 \longrightarrow BaSO_4 + 2\,HCl$
(d) $2\,Mg + O_2 \longrightarrow 2\,MgO$
(e) $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$.

34. A student has four unlabelled test tubes with HCl, NaOH, NaCl and Na_2CO_3 solutions. Suggest tests to identify each. 5

35. Explain, giving equations, the reaction of metals with 5
(a) oxygen
(b) water
(c) acids using one metal for each.

36. Explain tetravalency, catenation and covalent bonding of carbon with three reasons for the large number of carbon compounds. 5

OR

Describe an experiment to show that sunlight is necessary for photosynthesis or that CO_2 is needed, with the observations. 5

37. Trace the path of an impulse when you step on a thorn: describe receptors, neurons, brain, response and why the foot moves before pain is felt. 5

OR

Describe different methods of asexual reproduction in unicellular and multicellular organisms. State why they are not used by complex animals. 5

38. In pea plants, round seed (R) is dominant to wrinkled (r), and yellow (Y) to green (y). A plant RrYy is self-pollinated. Make a Punnett square outline: list gametes, give the F2 phenotypes and the ratio. 5

OR

An object 3 cm high is placed 20 cm from a concave mirror of focal length 12 cm. Find the position, nature, size of the image and magnification. Draw the ray-diagram rules used. 5

END OF QUESTION PAPER **Total: 80 Marks**

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